Automated Evaluation of Handwritten Entity Relationship Diagrams Using Object Detection, OCR, and Large Language Models.

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Abstract

This paper presents the design and preliminary evaluation of an automated system for assessing Entity Relationship (ER) diagrams from image input. The proposed Diagram Evaluation Engine combines object detection, optical character recognition (OCR), image processing, structural reconstruction, and rubric-based grading. A YOLOv8 model detects entities, weak entities, relationships, attributes, and primary keys from diagram images. OCR is then applied to detected image crops to recover diagram labels, and OpenCV-based edge and line detection identifies connections between components. The detected objects and connections are converted into a structured JSON representation of entities, attributes, and relationships, which is evaluated against a configurable marking guideline by a locally hosted large language model. The system outputs annotated images, structured JSON files, OCR results, detected connections, criterion-level marks, and feedback. Preliminary evaluation of the object detection component on a small, labeled dataset achieved 88.4% precision and 61.3% mAP@0.5. Evaluation of OCR accuracy, structural reconstruction, and grading agreement with expert marks is identified as ongoing work. The proposed pipeline supports scalable and consistent assessment of student ER diagrams while reducing manual marking effort.

Keywords— ER Diagram, YOLOv8, computer vision, educational technology, automated assessment

I. Introduction

Entity Relationship (ER) diagrams are widely used to represent entities, attributes, and relationships in database design. In educational settings, these diagrams are commonly assessed manually by instructors. Manual assessment is time consuming and may produce inconsistent results when many student submissions must be evaluated. The proposal underpinning this work identifies reliability, fairness, grading fatigue, and assessment workload as important concerns in teacher-based grading. It also identifies diagram-based answers such as ER diagrams as particularly demanding because they require interpretation of both textual labels and graphical structure. The proposed Diagram Evaluation Engine addresses this problem by automatically analyzing diagram images and transforming them into structured representations. The system identifies common ER diagram components, extracts their textual labels, reconstructs their connections, and evaluates the resulting structure against a configurable marking rubric. The resulting workflow is intended to reduce manual marking effort while providing consistent, criterion level marks and feedback.

II. Problem Statement

Research Reliable and fair assessment of student work is an important requirement in academic environments. However, teacher-based grading can be affected by subjective factors, differences in judgement, and grading practices, which may result in inconsistent assessment outcomes [1], [2]. Repetitive marking can further affect grading reliability because prolonged evaluation may lead to boredom and changes in the marks awarded to student work [3]. These challenges become more significant in computing education, where lecturers may need to evaluate a large number of diagram-based answers, including UML and Entity–Relationship (ER) diagrams, which require interpretation of both visual structures and their semantic meaning [4], [5].

Existing research has explored automated recognition and assessment of handwritten and diagram-based representations, but important limitations remain. Offline sketch recognition approaches have investigated the recognition of shapes and text from handwritten diagrams [6], while other approaches reconstruct strokes from static images before performing recognition [7]. However, stroke reconstruction can introduce errors, particularly when strokes intersect, overlap, or are affected by noise, which can reduce recognition accuracy [7]. Similarly, Arrow R-CNN has been investigated for handwritten diagram recognition, demonstrating the use of object-based

approaches for identifying diagram components [8]. Nevertheless, these existing approaches do not fully address the conditions of real examination answer sheets containing paper lines, rulers, background noise, and other visual artifacts.

Automated diagram assessment has also been investigated using digital diagram environments. Previous work has examined automated feedback for student diagrams, but students were required to create diagrams digitally rather than submit handwritten diagrams from examination answer sheets [9]. More recent research has investigated LLM-assisted automated evaluation of Entity–Relationship and schema diagrams, where students create diagrams using a graphical interface and the diagrams are converted into structured representations such as PlantUML for evaluation [5]. Other research on automated UML assessment has also examined machine learning and LLM-based grading approaches; however, these systems primarily operate on digitally generated diagrams rather than handwritten diagrams captured from real examination papers [4], [5].

Therefore, the key problem addressed by this research is the lack of an automated approach capable of extracting and evaluating handwritten ER diagrams directly from scanned or camera-captured examination answer sheets. Real examination papers introduce additional challenges such as paper lines, rulers, noise, overlapping diagram lines, handwritten labels, and variations in valid diagram structures. Existing approaches mainly focus on clean digital diagrams, stroke reconstruction, or digitally structured diagram representations [4]– [9]. As a result, there is a need for a system that can detect handwritten ER diagram components, extract their labels, identify their connections, reconstruct the diagram structure, and evaluate the resulting representation against a marking guideline. The proposed research addresses this problem by combining object detection, OCR, image processing, structural reconstruction, and automated rubric-based grading to support more consistent and efficient assessment of handwritten ER diagrams.

III. Research Gap

The research proposal identifies a gap between automated diagram assessment systems that operate on digitally structured diagrams and the practical requirement to assess diagrams drawn by students on real examination answer sheets. The expected paper background is therefore centered on an image-based pipeline that starts from the submitted diagram image rather than from an already structured UML or ER representation.

IV. System Overview And Methodology

The proposed system receives an image of an Entity Relationship diagram as input. The image passes through a sequence of processing stages: diagram object detection, image crop generation, OCR preprocessing and label extraction, line and connection detection, ER structure reconstruction, rubric based automated grading, and result storage and presentation.

The processing engine is implemented as a Python based system and is integrated with a web server endpoint. The endpoint accepts an image and an optional evaluation guideline. The final response contains detected objects, OCR results, connections, reconstructed structure, annotated image, marks, and feedback.

A. Detection Model

The object detection stage uses the Ultralytics YOLOv8 framework. The custom dataset contains five object classes: Attributes, Entities, Primary Key, Relationships, and Weak Entity. YOLOv8 is a single stage object detection framework that supports training, validation, inference, and related computer vision tasks. The configuration used by the proposed system is based on the YOLOv8 large detection model.

Parameter	Configured value
Model	YOLOv8 large (YOLOv8l)
Image size	640 pixels
Training epochs	100
Batch size	8
Number of classes	5
Inference confidence	0.60

Table 1 Model Details

For each detected object, the system records the class label, bounding box, confidence score, identifier, and crop path. The detection stage also saves cropped regions and text prediction files for later processing. The five classes are treated as the visual building blocks required by the later reconstruction stage.

B. OCR Processing

After object detection, the detected image regions are passed to the OCR module. The OCR pipeline improves the quality of small diagram labels through image preprocessing before the text extraction request is made. The processed crop is sent to the OCR.Space API using OCR Engine 3. The returned text is associated with the corresponding detected object. OCR results are stored in JSON, CSV, and text formats. The OCR module also includes result caching based on the SHA-256 digest of the processed image, reducing repeated requests when identical image crops occur.

C. Connection Detection

The system uses OpenCV to identify lines in the original diagram image. Canny edge detection is first applied to create an edge image. Probabilistic Hough line detection is then used to identify line segments. For every detected line, the endpoints are matched to the nearest detected objects using bounding box centers. Connections are represented using the identifiers of the two associated objects together with the coordinates of the detected line. Duplicate connections are removed, and lines whose endpoints map to the same object are ignored. This method provides a practical approximation for reconstructing diagram connectivity from raster images.

The connection method has known limitations. Highly curved lines, crossing lines, disconnected lines, and incorrectly positioned endpoints may reduce reconstruction

accuracy. Consequently, connection detection is evaluated separately from object detection and OCR.

D. ER Structure Reconstruction

The detected objects and connections are converted into a structured Entity Relationship representation. The system treats Entities and Weak Entity as entity types. Attributes and Primary Key are treated as attribute types, while Relationships are represented using the Relationships class.

```
{
  "entities": {
    "entity_name": {
      "attributes": ["attribute_name"]
    }
  },
  "relationships": [
    {
      "name": "relationship_name",
      "entities": ["entity_name"],
      "attributes": ["attribute_name"]
    }
  ]
}
```

Figure 1 Structure JSON

E. Rubric Based Grading

The grading module evaluates the reconstructed structure against a configurable marking guideline. The guideline contains criteria, criterion identifiers, descriptions, and maximum marks. The extracted structure and guideline are sent to a locally hosted LLM model.

The grading instruction is constrained to the extracted diagram structure. The model is instructed to avoid inferring missing objects or connections, return one result for every rubric criterion, assign zero marks when required information is missing or unverifiable, respect the maximum mark for each criterion, and return structured JSON containing marks and feedback.

V. Experimental Methodology

The system is to be evaluated using a labeled dataset of Entity Relationship diagram images. The dataset is intended to contain diagrams with different layouts, image qualities, font sizes, object counts, and connection patterns. The evaluation is performed at both component level and complete pipeline level.

A. Object Detection Performance

The five classes—attributes, primary keys, entities, weak entities, and relationships—will be evaluated independently because each class may have different visual characteristics.

B. OCR Performance

OCR performance will be evaluated using character accuracy, word accuracy, exact label matching accuracy, and OCR failure rate. These metrics measure how accurately the system extracts text labels from detected ER diagram components.

C. Structural Reconstruction Performance

Structural reconstruction will be evaluated using entity identification accuracy, attribute association accuracy, relationship identification accuracy, connection detection accuracy, and unmatched connection rate. These metrics measure how accurately the system reconstructs the ER diagram structure from detected components and connections.

D. Grading Performance

Grading performance will be evaluated using agreement with expert marks, mean absolute error, criterion-level accuracy, classification accuracy, and feedback consistency. These metrics measure how closely the automated grading results match expert evaluation and how consistently the system provides grading and feedback.

Metric	Status
Number of training images	350
Number of validation images	5
Object detection precision	88.4%
Object detection mAP	61.3% (mAP@0.5)
Average processing time	37.1 ms/image

Figure 2 Model Evaluation Table

VI. Experimental Setup

The experiments were conducted to train and evaluate a custom YOLOv8 object detection model for identifying components of Entity Relationship (ER) diagrams. Different ER diagram images were collected and manually labeled to create the dataset used for model training. The diagrams included variations in layout, component positioning, and labeling to provide different visual examples for the object detection model.

The dataset was annotated according to five ER diagram component classes: Attributes, Entities, Primary Keys, Relationships, and Weak Entities. Each relevant component in the ER diagram images was assigned to its corresponding class during the labeling process. The custom YOLOv8 model was trained using Google Colab. The training environment provided GPU-based computation for model training. The YOLOv8 Large model was configured for the five custom classes and trained using the prepared labeled ER diagram dataset.

The main training configuration used in the experiment was an image size of 640 pixels, a batch size of 8, and 100 training epochs. The model was configured with five detection classes corresponding to the annotated ER diagram components. After training, the custom YOLOv8 model was used to detect ER diagram components from previously unseen diagram images. The detection results were evaluated using standard object detection metrics, including precision, recall, F1 score, mean Average Precision (mAP), and per-class detection accuracy. The performance of each of the five classes was considered separately because the visual characteristics of attributes, entities, primary keys, relationships, and weak entities differ.

The trained model was subsequently used as the object detection component of the proposed ER diagram evaluation pipeline. Its detected objects provide the input

for the subsequent OCR, connection detection, and structural reconstruction stages.

VII. Preliminary Experimental Results

At the current stage, evaluation has been completed only for the object detection component of the pipeline; OCR, connection detection, structural reconstruction, and grading performance (as defined by the metrics in Section V) have not yet been measured and are reported as future work (Section X).

The custom YOLOv8l model was trained to detect five ER diagram component classes: Attributes, Entities, Primary Keys, Relationships, and Weak Entities, using 350 training images and 100 epochs at an image size of 640 pixels and a batch size of 8. The model was evaluated on a held-out validation set of 5 images, achieving an object detection precision of 88.4% and a mAP@0.5 of 61.3%, with an average inference time of 37.1 ms per image (Table 1).

We note the validation set size (n = 5) is too small to support strong generalization claims; these figures should be treated as an early sanity check on the detection model rather than a representative performance estimate**.** A properly sized held-out test set is required before these numbers can be reported as results (see Section IX, Limitations).

Per-class detection performance, precision/recall curves, and loss curves from training are shown in Figure 3. Figure 4 shows the distribution of predicted labels across the five classes on the validation set, and Figure 5 visualizes detection accuracy across a sample diagram, illustrating cases of correct and missed component detections.

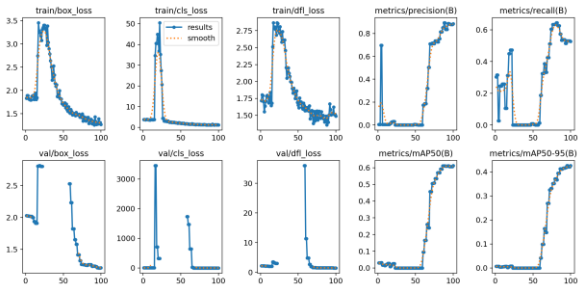


Figure 3 Key Final Results

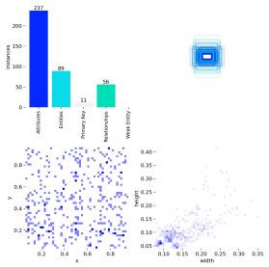


Figure 4 Label Distribution Result

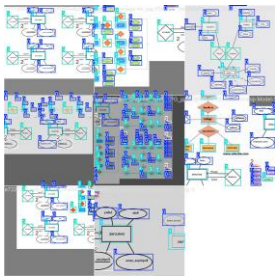


Figure 5 Accuracy Visualization

VIII. Discussion

The pipeline integrates visual component identification, OCR based label extraction, line detection, structural reconstruction, and rubric based grading. The output of one stage becomes the input of the next stage. This creates a dependency chain in which an upstream recognition error can affect downstream outputs. Incorrect object detection can cause OCR to process the wrong region. OCR errors can produce incorrect entity or relationship names. Incorrect line detection can produce missing or false connections. These errors may then affect the final rubric-based grade.

For this reason, the system should be evaluated both as a complete pipeline and as separate components. Component level evaluation helps determine whether errors originate from object detection, OCR, connection detection, structural reconstruction, or language model grading. This separation also makes the failure analysis more directly connected to the processing stage that produced the error.

IX. Limitations

The connection detection algorithm assigns line endpoints to the nearest detected object. Its accuracy may decrease when multiple objects are positioned close to each other.

The OCR component relies on the external OCR.Space service and requires an API key. Therefore, the OCR process depends on the availability of external service. OCR performance may decrease for very small, handwritten, blurred, or overlapping labels. This can affect the accuracy of the extracted diagram information. The language model grading result depends on the quality of the extracted ER diagram structure. Errors in earlier processing stages may therefore affect the final grading result. The system does not infer information that is not explicitly supported by the extracted data. Consequently, incomplete or unclear extracted information may limit the grading process.

The reported model performance requires validation using an independent test dataset to assess the model's generalization capability. The current system focuses on the supported Entity Relationship diagram classes. Therefore, it may not generalize directly to other diagram notations without additional training and adaptation.

X. Future Work

Future work will focus on training the model using a larger and more diverse dataset of Entity Relationship

diagrams. This can improve the robustness of the system across different diagram layouts and drawing styles. The system can be extended to support cardinality and participation constraints. This would enable more comprehensive analysis of Entity Relationship diagrams. The line association process can be improved using graph-based spatial reasoning. This may provide more accurate associations between detected objects and their corresponding connections. Confidence-aware structure reconstruction can be introduced to identify uncertain detections and connections. This can help improve the reliability of the reconstructed diagram structure. Future work can also compare multiple language models for rubric-based grading. This would allow the grading performance of different models to be evaluated and compared.

A human-in-the-loop correction mechanism can be introduced for uncertain detections. This would allow users to review and correct detection or reconstruction errors before final grading. The system can be further evaluated for fairness and consistency across different diagram drawing styles. This can help determine whether variations in drawing style affect the system's performance.

Finally, a benchmark dataset for automated Entity Relationship diagram assessment can be developed. Such a dataset would provide a standardized basis for evaluating and comparing future automated assessment approaches.

XI. Conclusion

Th**is** paper presents the architecture and preliminary evaluation of an automated Diagram Evaluation Engine for analyzing and grading Entity Relationship diagrams. The system combines YOLOv8 object detection, OCR, OpenCV-based image processing, structural reconstruction, and local large language model grading, producing both visual and machine-readable evaluation outputs — detected objects, diagram labels, connections, reconstructed structures, criterion-level marks, and feedback.

Preliminary results on the object detection component (88.4% precision, 61.3% mAP@0.5) suggest the approach is technically feasible, though evaluation on a larger, independent test set is required to establish reliable performance figures. Full evaluation of OCR accuracy, structural reconstruction, and grading agreement with expert markers — including comparison against human-graded diagrams — remains for future work.

The proposed pipeline provides a foundation for scalable, consistent assessment of handwritten ER diagrams in educational settings, addressing a gap between existing diagram-assessment systems (which typically assume digitally structured input) and the practical need to assess diagrams drawn on paper. Its effectiveness as a complete system will be established through the controlled, component-level and pipeline-level evaluation outlined in Section V.

XII. References

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